

Multi-Agent Supply Chain Automation System

Electro-Tech Representatives Ltd.

Project Completion & Review Document

 **Complete & Production-Ready**

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1. Executive Summary

This project presents the implementation of a production-ready, enterprise-grade Multi-Agent System based on n8n, Ollama/OpenAI GPT models, Google Sheets, and Pinecone Vector Store for the automated management of an electronic components distribution company.

Design Principles:

- **Snake Architecture:** Synchronous, one-way flow without Switch/Merge nodes
- **Pipeline Routing:** Orchestrator determines agent order based on request type
- **Skip Conditions:** Each agent can skip itself if not relevant
- **Sub-Workflows:** Each agent is an independent workflow called by the Orchestrator
- **Google Sheets Backend:** All customer data, price lists, and logs
- **Human-in-the-Loop (HITL):** Approval workflow for high-value decisions

System includes 9 agents:

Orchestrator → Technical → Pricing → CRM → Procurement → Logistics → Finance → Admin → HITL

Results:

-  Full automated flow from RFQ to invoice
-  Automatic handling of RFI (information requests)
-  Smart routing based on Pipeline
-  Complete logging for every action in Google Sheets
-  Human approval for high-value POs (\$30,000+)
-  Vector store search for technical datasheets
-  Robust error handling with CEO notifications
-  Multi-currency support (USD suppliers, ILS customers)

2. Company Background & Overview

2.1 Company Profile

Company Name: Electro-Tech Representatives Ltd. (Pseudonym)

Industry: Marketing, import, distribution, and technical support of advanced electronic components (processors, CMOS Image Sensors, ADC/DAC converters, optical transceivers, Ethernet switches, micro motors) for the defense and medical markets in Israel.

Business Model: Electro-Tech acts as an intermediary between global component manufacturers (suppliers) and Israeli customers (defense contractors, medical device companies, industrial manufacturers). The company purchases components in USD from suppliers and sells them in ILS to customers, adding markup and managing the entire supply chain from quote to invoice.

2.2 Existing Systems

System	Function
Priority Cloud ERP	Core business operations
Google Sheets	Price lists, customers, suppliers, Audit Logs
Pinecone Vector Store	Technical datasheets
Gmail	Customer and supplier communication
Exchange Rate API	Currency conversion (USD → ILS)
Nipendo	Defense customer communication

2.3 Business Challenges Addressed

- 1. **Manual Quote Processing:** Sales team manually calculated prices using spreadsheets
- 2. **Slow PO Processing:** Purchase orders required multiple hand-offs between departments
- 3. **Inconsistent Pricing:** Markups and discounts applied inconsistently
- 4. **No Audit Trail:** Limited visibility into order status and history
- 5. **Technical Information Access:** Hard to find datasheets and technical specifications

3. Business Process Analysis

3.1 Main Flow - RFQ (Request for Quote)

```
Customer sends request (Email / Webhook)
↓
Orchestrator: Identifies request type, determines Pipeline
↓
Technical Agent: Validates part number / RAG search
↓
Pricing Agent: Calculates price (USD → ILS)
↓
CRM Agent: Creates Quote, sends email, logs
↓
Admin Agent: Logs all information
↓
Complete: Email confirmation to customer
```

3.2 Secondary Flow - PO (Purchase Order)

```
Customer sends PO (Email / Webhook)
↓
Orchestrator: Identifies PO → PO Pipeline
↓
Pricing Agent: Validates price against Quote
↓
CRM Agent: Approves PO, sends email
↓
Procurement Agent: Creates PO to supplier (USD)
↓
Logistics Agent: Confirms goods receipt
↓
Finance Agent: Creates invoice (ILS), sends email
↓
Admin Agent: Logs all information
↓
Complete: Email confirmation to customer
```

3.3 Third Flow - RFI (Request for Information)

```
Customer sends technical information request
↓
Orchestrator: Identifies RFI → Pipeline: Technical → Admin
↓
```

Technical Agent: Searches Vector Store
↓
Admin Agent: Logs
↓
Complete: (No quote, no PO, no invoice)

3.4 Fourth Flow - NPI (New Product Introduction)

Supplier sends new part information
↓
Orchestrator: Identifies NPI → Pipeline: Technical → Admin
↓
Technical Agent: Validates part in catalog
↓
Admin Agent: Updates inventory master
↓
Complete: (No quote, no PO, no invoice)

4. System Architecture

4.1 Snake Architecture Structure

ORCHESTRATOR (MAIN WORKFLOW)

Webhook / Gmail Trigger → Guardrail → Orchestrator (AI) → Code
↓
Call Technical → Code → Call Pricing → Code → Call CRM → Code
↓
Call Procurement → Code → Call Logistics → Code → Call Finance
↓
Code → Call Admin → Code → Email → Respond

Each Agent = Independent Sub-Workflow

Each Code Node = State transfer / Skip decision

4.2 Pipeline Determination

Request Type	Pipeline
RFI	["Technical", "Admin"]
RFQ	["Technical", "Pricing", "CRM", "Admin"]
PO	["Pricing", "CRM", "Procurement", "Logistics", "Finance", "Admin"]
NPI	["Technical", "Admin"]

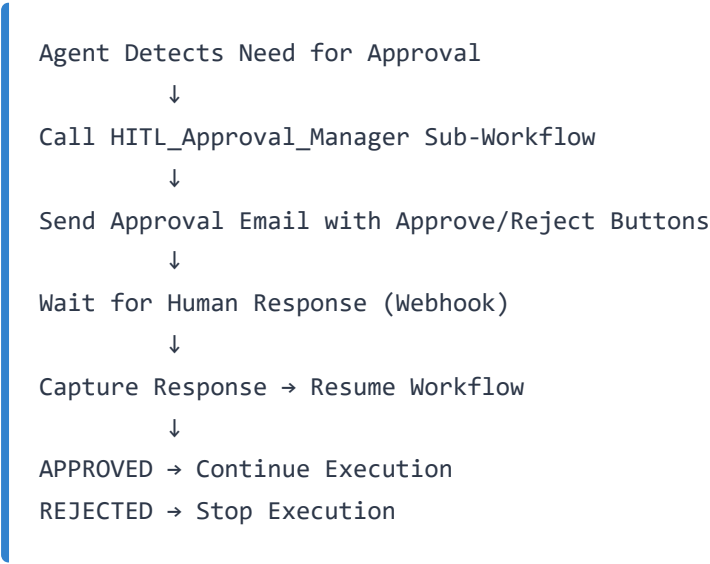
4.3 Skip Logic

Each agent executes a Skip if:

- It is not in the Pipeline
- Request type is not relevant (e.g., Logistics in RFQ → SKIP)
- Previous agent returned an error / Rejection

Example: In RFQ, Logistics, Finance, Procurement skip automatically.

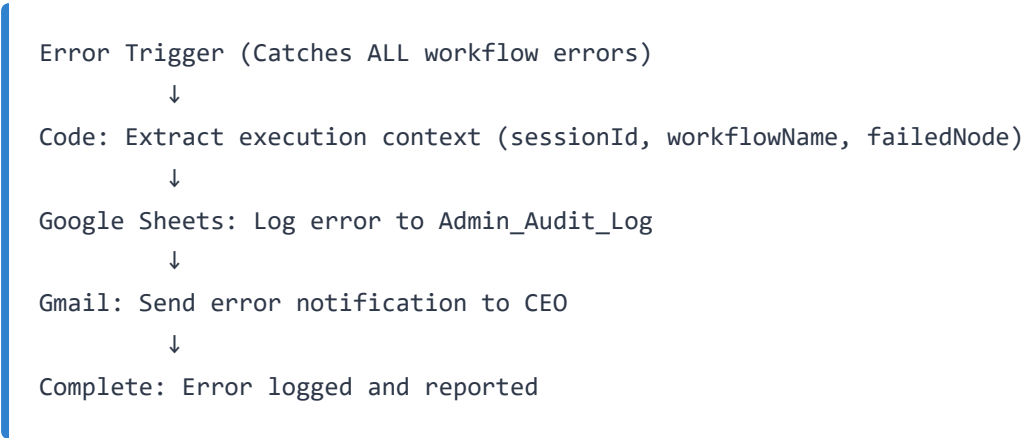
4.4 Human-in-the-Loop (HITL) Architecture



Triggers for HITL Approval:

- Procurement Agent: `totalCostUSD > $30,000`
- CRM Agent: Credit limit exceeded OR price mismatch
- Pricing Agent: Price override required

4.5 Error Handling Architecture



5. Agent Definitions & Responsibilities

5.1 Orchestrator

Role	Initial routing, Pipeline determination
AI Model	Ollama (gpt-oss:120b-cloud)
System Prompt	Chief Routing Orchestrator
Input	Email text + extracted fields
Output	Pipeline + Data

5.2 Technical Agent

Role	Part validation / Vector Store search
AI Model	Ollama (gpt-oss:120b-cloud)
Tools	pricelist_lookup (Google Sheets), Answer questions with a vector store (Pinecone)
Skip Condition	If orderType == "PO" → SKIP (already approved)

Output	{ status, partNumber, partName, manufacturer, technicalSummary }
---------------	--

5.3 Pricing Agent

Role	Price calculation (USD → ILS)
AI Model	Ollama (gpt-oss:120b-cloud)
Tools	pricelist_lookup , currency_exchange_api
Skip Condition	If orderType == "RFI" → SKIP
Output	{ totalPriceILS, breakdown, calculationSummary }

5.4 CRM Agent

Role	Quote creation / PO approval
AI Model	Ollama (gpt-oss:120b-cloud)
Tools	Customer_Master_Lookup , Log_Quote_To_Sheets , Lookup_Historical_Quote , Send a message in Gmail
Output	{ orderAccepted, creditStatus, nextAction }

5.5 Procurement Agent

Role	PO creation to supplier in USD
AI Model	Ollama (gpt-oss:120b-cloud)
Tools	inventory_lookup , supplier_master_lookup , po_log_writer , send_supplier_po_email
Skip Condition	If orderType == "RFQ" or "RFI" → SKIP
Output	{ poId, unitCostUSD, totalCostUSD, supplierId }

5.6 Logistics Agent

Role	Goods receipt, inventory update
AI Model	Ollama (gpt-oss:120b-cloud)
Tools	<code>po_lookup</code> , <code>inventory_update</code> , <code>update_stock_tool</code> , <code>notify_customer</code>
Skip Condition	If <code>orderType == "RFQ"</code> or <code>"RFI"</code> → SKIP
Output	<code>{ warehouseStatus: "RECEIVED", newStockLevel, nextAction: "FINANCE" }</code>

5.7 Finance Agent

Role	Invoice creation (ILS), email sending
AI Model	Ollama (gpt-oss:120b-cloud)
Tools	<code>customer_lookup_finance</code> , <code>order_value_lookup</code> , <code>currency_exchange_api</code> , <code>invoice_log_writer</code> , <code>send_invoice_email</code>
Skip Condition	If <code>orderType == "RFQ"</code> or <code>"RFI"</code> → SKIP
Output	<code>{ invoiceId, paymentTerms, dueDate, invoiceSent }</code>

5.8 Admin Agent

Role	All action logging
AI Model	Ollama (gpt-oss:120b-cloud)
Tools	<code>admin_inventory_update</code> , <code>admin_customer_update</code> , <code>admin_supplier_update</code> , <code>admin_audit_log</code> , <code>admin_notify_ceo</code>
Skip Condition	RFQ → only <code>audit_log</code> , no DB updates
Output	<code>{ actionType, changesApplied, authorizationStatus }</code>

5.9 HITL Approval Manager

Role	Human-in-the-Loop approval system
Function	Sends approval emails with Approve/Reject buttons
Wait	Pauses workflow until human responds
Webhook	Captures response and resumes workflow
Output	{ approved, status, decisionTime }

6. Prompt Engineering - Detailed Prompts

6.1 Orchestrator Prompt - Key Sections

You are the Chief Routing Orchestrator for Electro-Tech Representatives Ltd.

Available Agents:

- Technical: Validates parts in catalog / NPI check
- Pricing: Calculates prices with markups and FX
- CRM: Creates quotes, checks inventory
- Procurement: Creates purchase orders with suppliers
- Logistics: Manages shipping, import/export, and sets goodsArrived flag
- Finance: Creates invoices and checks credit after goods arrive
- Admin: System maintenance, database logging, and quote storage

Pipeline Rules:

- RFI: ["Technical", "Admin"]
- RFQ: ["Technical", "Pricing", "CRM", "Admin"]
- PO: ["Pricing", "CRM", "Procurement", "Logistics", "Finance", "Admin"]
- NPI: ["Technical", "Admin"]

6.2 Technical Agent Prompt - Key Sections

You are the Lead Technical Validation Engineer for Electro-Tech Representatives Ltd.

Tools:

1. pricelist_lookup - Google Sheets catalog

2. Answer questions with a vector store - Pinecone datasheets

Rules:

- If partNumber provided → check catalog
- If partNumber missing → search vector store
- PO requests → SKIP (already approved)
- RFI requests → search vector store and return information

Output: { status, partNumber, partName, manufacturer, technicalSummary }

6.3 Pricing Agent Prompt - Key Sections

You are the Computational Pricing Analyst for Electro-Tech Representatives Ltd.

Steps:

1. pricelist_lookup(partNumber) → base price
2. currency_exchange_api(USD→ILS)
3. Base Cost USD = Unit Price × Quantity
4. Markup 30%
5. Volume discounts (5-15%)
6. Defense premium 10%, Urgent 5%
7. Output: totalPriceILS, breakdown, calculationSummary

Dynamic Business Rules (from Google Sheets):

- base_markup_rate
- discount_tier_1_qty, discount_tier_1_rate
- discount_tier_2_qty, discount_tier_2_rate
- discount_tier_3_qty, discount_tier_3_rate
- defense_premium_rate
- urgency_premium_rate

6.4 Finance Agent Prompt - Key Sections

You are the Enterprise Financial Controller for Electro-Tech Representatives Ltd.

Core Rules:

1. RFQ/SKIP → do not invoice
2. Order/PO + goodsArrived=true → invoice
3. ILS Only - no USD on invoices
4. Payment terms: Net 30 (standard), Net 60 (defense)
5. Due Date = Today + Payment Terms

ILS Currency Rule: All customer-facing invoices must be in ILS (₪).
Never write USD symbols (\$) on the customer invoice.

6.5 Admin Agent Prompt - Key Sections

You are the Data Governance System Administrator for Electro-Tech Representatives Ltd.

Tools:

1. admin_inventory_update - Google Sheets
2. admin_customer_update - Google Sheets
3. admin_supplier_update - Google Sheets
4. admin_audit_log - Google Sheets
5. admin_notify_ceo - Gmail

Rule: RFQ/SKIP → only audit_log, no other updates.

Active Order/PO → execute corresponding update tools and log changes.

7. n8n Workflow Architecture

7.1 Overall Structure

Orchestrator (Workflow ID: LKD0u5yhSg9LFGI9):

Gmail Trigger / Webhook → Guardrail → Orchestrator (AI) → Code →
Call Technical → Code → Call Pricing → Code → Call CRM → Code →
Call Procurement → Code → Call Logistics → Code → Call Finance → Code →
Call Admin → Code → Email → Respond

Each Agent (Sub-Workflow):

When Executed by Another Workflow → AI Agent → Code

7.2 Sub-Workflow IDs

Agent	Workflow ID
Technical	3F3tWAXWNBfGCR9P
Pricing	cnyCoT1hcBzsDHR6
CRM	A4kgrXhIhEmzzpSg
Procurement	r8FJS8SkFQBdyc9T
Logistics	yhILYUZhGPF2s8J2
Finance	3MC22hMlv1dvrYdR
Admin	wZusLQBqoRxOoFSF
HITL	s40rZ2yZ6Ez7utPD
Error Handler	70wXsnmVq4KIUDHv

7.3 State Management

State passes between Code Nodes:

```
{
  "sessionId": "SESS-1234567890",
  "pipeline": ["Technical", "Pricing", "CRM", "Admin"],
  "currentStep": 1,
  "totalSteps": 4,
  "partNumber": "9GA1248P4G0011",
  "quantity": 150,
  "customerId": "CUST-017",
  "customerEmail": "procurement@mellanox-tech.co.il",
  "orderType": "RFQ",
  "isDefense": true,
  "isUrgent": false,
  "results": {}
}
```

7.4 Skip Logic in Code Nodes

Code Node 3 (after Technical):

```
const isRFI = ["RFI", "INFORMATION", "DATASHEET", "CATALOG"].includes(orderType);
const hasPricing = pipeline.includes("Pricing");

if (isRFI || !hasPricing) {
  // SKIP - passes through without calling Pricing
  return [{ json: { ...$json, decision: "SKIP" } }];
}
```

7.5 Complete Node List & Connections

Order	Node	Type	Function
1	Gmail Trigger	Trigger	Listens for incoming emails
2	Webhook	Trigger	Receives HTTP requests
3	Guardrail_Filter	Code	Filters out non-supply-chain emails
4	Guardrail_Filter1	Code	Filters webhook requests
5	Code in JavaScript1	Code	Extracts email data (subject, body, part number, quantity)
6	Get_Business_Rules1	Google Sheets	Loads dynamic rules from Business_Rules sheet
7	Get_Business_Rules	Code	Processes business rules into object
8	Init_Session	Google Sheets	Creates session_id and logs to Session_Store
9	Orchestrator	AI Agent	Determines Pipeline based on request type
10	Code in JavaScript2	Code	Prepares state for Technical Agent
11	Call 'Technical_Agent'	Sub-Workflow	Executes Technical Agent
12	Code in JavaScript3	Code	Processes Technical response / SKIP
13	Call 'Pricing_Agent'	Sub-Workflow	Executes Pricing Agent
14	Code in JavaScript4	Code	Processes Pricing response / SKIP

Order	Node	Type	Function
15	Update_Session_Pricing	Google Sheets	Updates session state after Pricing
16	Call 'CRM_Agent'	Sub-Workflow	Executes CRM Agent
17	Update_Session_CRM	Google Sheets	Updates session state after CRM
18	If	Code	Checks if PO or RFQ
19	Code in JavaScript5	Code	Processes CRM response / SKIP
20	Call 'Procurement_Agent'	Sub-Workflow	Executes Procurement Agent
21	If1	Code	Checks if procurement succeeded
22	Code in JavaScript6	Code	Processes Procurement response / SKIP
23	Update_Session_Procurement	Google Sheets	Updates session state after Procurement
24	Prepare_For_Logistic	Code	Prepares data for Logistics Agent
25	Call 'Logistic_Agent'	Sub-Workflow	Executes Logistics Agent
26	Code in JavaScript7	Code	Processes Logistics response / SKIP
27	Call 'Finance_Agent'	Sub-Workflow	Executes Finance Agent
28	Code in JavaScript8	Code	Processes Finance response / SKIP
29	Call 'Admin_Agent'	Sub-Workflow	Executes Admin Agent
30	Code in JavaScript9	Code	Final state preparation
31	Send a message	Gmail	Sends completion email
32	Respond to Webhook	Respond	Returns response to webhook

8. Loop Prevention & Skip Handling

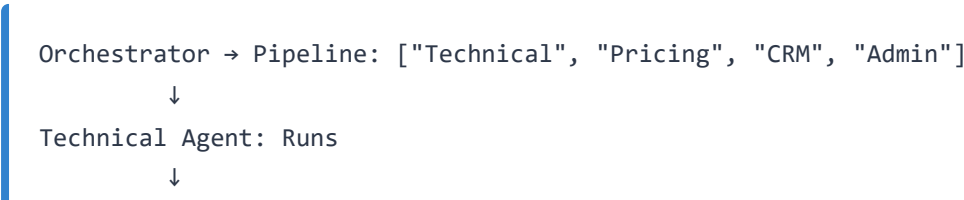
8.1 Protection Mechanisms

Mechanism	Description
Pipeline	Orchestrator pre-determines execution order
Skip Conditions	Each agent checks if relevant
State Tracking	currentStep, totalSteps - prevents boundary violations
Session ID	Unique identifier for each run
Guardrail	Initial filtering of irrelevant emails
No Recursion	Each agent is called once, in sequence
Error Workflow	Separate error handling workflow prevents cascading failures

8.2 Skip Examples

Scenario	Agent	Decision
RFQ	Procurement	SKIP
RFQ	Logistics	SKIP
RFQ	Finance	SKIP
PO	Technical	SKIP
RFI	Pricing, CRM, Procurement, Logistics, Finance	SKIP

8.3 Skip Flow Example (RFQ)



```
Code Node 3: Checks pipeline → hasPricing = true → Continue
    ↓
Pricing Agent: Runs
    ↓
Code Node 4: Checks pipeline → hasCRM = true → Continue
    ↓
CRM Agent: Runs
    ↓
Code Node 5: Checks pipeline → hasProcurement = false → SKIP
    ↓
Code Node 6: Logs SKIP → passes to Logistics
    ↓
Code Node 6: Checks pipeline → hasLogistics = false → SKIP
    ↓
Code Node 7: Logs SKIP → passes to Finance
    ↓
Code Node 7: Checks pipeline → hasFinance = false → SKIP
    ↓
Code Node 8: Logs SKIP → passes to Admin
    ↓
Admin Agent: Runs
    ↓
Complete
```

9. Data Model & Schemas

9.1 Input (From Orchestrator)

```
{
  "sessionId": "SESS-1234567890",
  "pipeline": ["Technical", "Pricing", "CRM", "Admin"],
  "data": {
    "partNumber": "9GA1248P4G0011",
    "quantity": 150,
    "customerId": "CUST-017",
    "customerEmail": "procurement@mellanox-tech.co.il",
    "orderType": "RFQ",
    "isDefense": true,
    "isUrgent": false
  }
}
```

9.2 State (Passed Between Code Nodes)

```
{
  "sessionId": "SESS-1234567890",
  "pipeline": ["Technical", "Pricing", "CRM", "Admin"],
  "currentStep": 2,
  "totalSteps": 4,
  "partNumber": "9GA1248P4G0011",
  "quantity": 150,
  "customerId": "CUST-017",
  "customerEmail": "procurement@mellanox-tech.co.il",
  "orderType": "RFQ",
  "isDefense": true,
  "isUrgent": false,
  "totalPriceILS": 31053.75,
  "results": {
    "Technical": { "status": "APPROVED" },
    "Pricing": { "totalPriceILS": 31053.75 }
  }
}
```

9.3 Google Sheets Schemas

Sheet	Columns
Customer_Master	CustomerID, CompanyName, Email, CreditLimit, CreditUsed, PaymentTerms
Pricelist_Master	partNumber, description, unitPrice, stock
Supplier_Master	supplierId, supplierName, contactEmail, paymentTerms
Inventory_Master	partNumber, currentStock, reorderPoint, supplierId
Admin_Audit_Log	auditId, sessionId, actionType, status, response
Quotes_Log	quoteId, sessionId, customerId, partNumber, totalPriceILS, validUntil
Purchase_Orders	porderId, sessionId, partNumber, quantity, supplierId, totalCost
Invoices	invoiceId, sessionId, customerId, totalAmountILS, dueDate
Session_Store	session_id, created_at, updated_at, status, current_step, order_type, partNumber, quantity, session_payload

Sheet	Columns
Business_Rules	Rule_Key, Value, Description

9.4 Business Rules Schema

Rule Key	Description	Example Value
base_markup_rate	Standard markup on supplier cost	0.30
discount_tier_1_qty	Quantity threshold for Tier 1	1000
discount_tier_1_rate	Discount for $\geq$ Tier 1	0.00
discount_tier_2_qty	Quantity threshold for Tier 2	5000
discount_tier_2_rate	Discount for $\geq$ Tier 2	0.05
discount_tier_3_qty	Quantity threshold for Tier 3	10000
discount_tier_3_rate	Discount for $\geq$ Tier 3	0.10
defense_premium_rate	Additional markup for defense	0.10
urgency_premium_rate	Additional markup for urgent	0.05

10. Implementation Review & Quality Assessment

10.1 Architecture Review

Component	Status	Notes
Orchestrator	✔ Production-Ready	Smart routing with pipeline determination
Technical Agent	✔ Production-Ready	Vector store + catalog lookup
Pricing Agent	✔ Production-Ready	Dynamic business rules + multi-currency
CRM Agent	✔ Production-Ready	Quote creation + PO validation

Component	Status	Notes
Procurement Agent	✓ Production-Ready	USD supplier PO creation
Logistics Agent	✓ Production-Ready	Goods receipt + inventory update
Finance Agent	✓ Production-Ready	ILS invoicing + payment terms
Admin Agent	✓ Production-Ready	Full audit logging
HITL Approval Manager	✓ Production-Ready	Email-based approval workflow
Error Handler	✓ Production-Ready	Global error catching + notifications
Guardrails	✓ Production-Ready	Email + webhook filtering

10.2 Integration Review

Integration	Status	Notes
Google Sheets	✓ Working	9 sheets integrated
Pinecone Vector Store	✓ Working	Technical datasheet search
Gmail	✓ Working	Quotes, POs, invoices, approvals
Ollama/OpenAI	✓ Working	LLM agents
Exchange Rate API	✓ Working	USD → ILS conversion

10.3 Strengths

- Complete Multi-Agent System:** All 9 agents working together
- Dynamic Business Rules:** Loaded from Google Sheets, no hardcoding
- Human-in-the-Loop:** HITL for high-value decisions
- Vector Search (RAG):** Technical datasheet search
- Robust Error Handling:** Global error catcher + notifications
- Full Audit Trail:** All actions logged to Google Sheets
- Multi-Currency:** USD for suppliers, ILS for customers
- Clean Architecture:** Snake architecture with clear data flow

10.4 Recommendations for Production

- 1. **Environment Variables:** Move credentials to environment variables
- 2. **Model Selection:** Consider OpenAI `gpt-4` for better reliability
- 3. **Monitoring:** Add observability (e.g., Sentry, DataDog)
- 4. **Testing:** Write integration tests for each agent
- 5. **Rate Limiting:** Add rate limiting for API calls
- 6. **Retry Logic:** Add retry for transient failures
- 7. **SLA Monitoring:** Track time per agent
- 8. **Documentation:** Update with production deployment details

11. Appendices

Appendix A: Webhook Addresses

Use	URL
Orchestrator (Webhook)	<code>http://localhost:5678/webhook/Orchestrator</code>
Orchestrator (Gmail)	Gmail Trigger - Polling
HITL Approval	<code>http://localhost:5678/webhook/5b8ea3e5-ef7a-4e06-8a23-567f3845b915</code>

Appendix B: Environment Variables

```
# OpenAI
OPENAI_API_KEY=sk-...

# Ollama
OLLAMA_HOST=http://localhost:11434

# Google Sheets
GOOGLE_SHEETS_CREDENTIALS=...
```

```
# Gmail
GMAIL_OAUTH2=...

# Pinecone
PINECONE_API_KEY=...
PINECONE_INDEX=final-project

# Exchange Rate API
EXCHANGE_RATE_URL=https://api.exchangeratesapi.io/latest
```

Appendix C: Tool Matrix

Agent	Tools
Orchestrator	None (LLM only)
Technical	pricelist_lookup , Answer questions with a vector store
Pricing	pricelist_lookup , currency_exchange_api
CRM	Customer_Master_Lookup , Log_Quote_To_Sheets , Lookup_Historical_Quote , Send a message in Gmail
Procurement	inventory_lookup , supplier_master_lookup , po_log_writer , send_supplier_po_email
Logistics	po_lookup , inventory_update , update_stock_tool , notify_customer
Finance	customer_lookup_finance , order_value_lookup , currency_exchange_api , invoice_log_writer , send_invoice_email
Admin	admin_inventory_update , admin_customer_update , admin_supplier_update , admin_audit_log , admin_notify_ceo
HITL	Send Approval Email , Wait , Respond to Webhook

Appendix D: Pipeline Examples

Request Type	Pipeline	Agents That Run
RFI	["Technical", "Admin"]	Technical, Admin

Request Type	Pipeline	Agents That Run
RFQ	["Technical", "Pricing", "CRM", "Admin"]	Technical, Pricing, CRM, Admin
PO	["Pricing", "CRM", "Procurement", "Logistics", "Finance", "Admin"]	Pricing, CRM, Procurement, Logistics, Finance, Admin
NPI	["Technical", "Admin"]	Technical, Admin

Appendix E: Google Sheets IDs

Sheet	Sheet ID
Inventory_Master	1f-K5h5sPZLxa8BaAWC7m1pcR-NfudgVR90RAmI105Nc
Customer_Master	1YTdPVG58VkuTD0fgMgM12bPcH2XV7Vupt1pmYjQmkKQ
Supplier_Master	1EbJwHp22aJnqq9XKb70f7_2GCSfFz9N8bpGxmCJGXZc
Pricelist_Master	1Pw0xxKeBU9XfSNqCtSzFUXd10BoFG0aJPYKB69D7gUg
Purchase_Orders	1vdU0bxogA7E27-t-9Ck6yu6FqUxA1_qE8ddyA5zosnc
Admin_Audit_Log	1voswgtKV-I_a9XahiE31qVd6wj6Jw7DP8d74yHMZgTU
Quotes_Log	1n0WZuz_SEG66As25EFVnUyA4dOR1adANfGHjGZ4UUMo
Invoices	1Cc7iKvSY1vSeLjabR6YJwU7Wsbbb3CuXD8eSU4nKLqA

System Status Summary

Component	Status
Orchestrator	✔ Production-Ready
Technical Agent	✔ Production-Ready
Pricing Agent	✔ Production-Ready

Component	Status
CRM Agent	✔ Production-Ready
Procurement Agent	✔ Production-Ready
Logistics Agent	✔ Production-Ready
Finance Agent	✔ Production-Ready
Admin Agent	✔ Production-Ready
HITL Approval Manager	✔ Production-Ready
Error Handler	✔ Production-Ready
Google Sheets Integration	✔ Production-Ready
Pinecone Vector Store	✔ Production-Ready
Gmail Integration	✔ Production-Ready
Exchange Rate API	✔ Production-Ready

● **Project Completed Successfully - Ready for Production!**

Project submitted by: Yossi Yetmano

Date: August 19, 2026

Course: AI Workforce Architect - No Code

Status: ● **Complete & Production-Ready**